THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 9–12: **Light – Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** convex, object between F and O	Magnifying glass
2	**(c)** becomes parallel to the principal axis	Reversibility
3	**(c)** Incorrect — it is optically the densest; light is slowest	$v = c/n$
4	**(c)** pupil	Controlled by iris
5	**(b)** real and inverted	Convex lens
6	**(b)** concave lens	Diverging
7	**(a)** 6.25×10^{18}	$1/(1.6 \times 10^{-19})$
8	**(b)** ohm metre	Ω m
9	**(d)** 2 A	$I = Q/t$
10	**(b)** Student A is correct	Thumb along current
11	**(a)** perpendicular to the field	Sin 90°
12	**(a)** Correct	Two directions impossible

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	False — a convex lens is used; a concave lens always gives diminished images.	2
14	Loss of accommodation with age, due to weakening of ciliary muscles and reduced flexibility of eye lens.	2
15	It controls the size of the pupil and hence the amount of light entering the eye.	2
16	Very high melting point (3380 °C) and it emits light at high temperature.	2
17	Hold the conductor in the right hand with the thumb along the current; curled fingers give the direction of magnetic field lines.	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$\frac{\sin i}{\sin r}$ = constant for a given pair of media; $n_{ag} = 1/1.5 = 0.67$	2+1
19	Concave lens diverges rays from distant objects so that they appear to come from the far point of the myopic eye; image formed on the retina	3
20	Length × 2, area ÷ 2 ⇒ $R' = 4R = 40\ \Omega$	3
21	Use proper fuses/MCBs; avoid many appliances on one socket; use good insulation; repair damaged wires	3
22	(i) perpendicular to field (ii) parallel to field; Fleming's left-hand rule	1+1+1
	OR: Provides a low-resistance path for leakage current to earth, preventing electric shock; green wire	2+1
23	Torches/headlights (reflectors), shaving mirrors, dentist's mirrors, solar furnaces	3
	OR: Fastest in water (lowest n); densest: turpentine (highest n)	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) cannot see nearby objects; near point beyond 25 cm (ii) image formed behind retina; convex lens converges rays to form image on retina (iii) long focal length of eye lens; eyeball too small	1+3+1
	OR: (i) splitting of white light into colours (ii) different colours bend through different angles (refractive index differs) (iii) second inverted prism recombines the colours to give white light (iv) band of coloured components VIBGYOR	1+1+2+1
25	(i) aluminium rod suspended between poles of a horse-shoe magnet; when current flows, the rod is displaced (ii) the force reverses (iii) Fleming's left-hand rule	3+1+1

	OR: (i) thumb, forefinger, middle finger mutually perpendicular (ii) upward (towards top of page) (iii) loudspeakers, electric motor (fans)	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

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SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** 2.25×10^8 m/s	$3/1.33$
2	**(d)** −10 cm	$f = R/2$
3	**(d)** concave	Magnified image
4	**(c)** accommodation	Ciliary muscles
5	**(d)** hypermetropia	Far-sightedness
6	**(d)** ciliary muscles	Accommodation
7	**(a)** ampere	1 A = 1 C/s
8	**(a)** Correct	Resistivity
9	**(a)** 24 J	$W = QV$
10	**(c)** north pole and enter at the south pole	Outside the magnet
11	**(a)** uniform	Parallel lines
12	**(c)** green	Live — red/brown

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Due to refraction of light at the water–air surface.	2
14	Scattering of light by colloidal particles, making the path of light visible.	2
15	$I = V/R = 12/40 = 0.3$ A	2
16	False — they are connected in parallel so each gets the full voltage and can be switched independently.	2
17	The region around a magnet in which its magnetic force can be detected.	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$u = -10$, $f = +15$: $\frac{1}{v} = \frac{1}{15} + \frac{1}{10} = \frac{1}{6} \Rightarrow v = +6$ cm; $m = -v/u = 0.6$; virtual, erect, diminished	2+1
19	Far-sightedness — near point farther than 25 cm; causes: focal length of eye lens too long, eyeball too small; corrected by convex lens	1+1+1
20	(i) $R = V^2/P = 484\ \Omega$ (ii) $I = P/V = 0.45$ A (iii) $E = 0.1 \times 10 = 1$ kWh	1+1+1
21	Straight: concentric circles, weakens with distance; solenoid: like bar magnet, uniform inside, stronger with more turns	3
22	$R = 2f$; $R = -24$ cm	1+2
	OR: (i) ratio of speed of light in vacuum to that in medium (ii) central point of a lens through which light passes undeviated (iii) line through centres of curvature/optical centre	1+1+1
23	Due to atmospheric refraction, the Sun is visible about 2 minutes before actual sunrise and after actual sunset	3
	OR: Ray bends towards the base on entering and leaving; angle between the incident ray produced and emergent ray is the angle of deviation	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) the two refracting surfaces are parallel; the bending at both surfaces is equal and opposite (ii) lateral displacement (iii) decreases (iv) $v = c/n = 2 \times 10^8$ m/s	2+1+1+1
	OR: (i) change in direction of light passing from one medium to another (ii) $v = c/n = 2.25 \times 10^8$ m/s (iii) $n_{wg} = n_w/n_g = (4/3)/(3/2) = 8/9 \approx 0.89$	1+2+2
25	(i) $W = QV = (It)(IR) = I^2Rt$ (ii) $H = 16 \times 50 \times 60 = 48\,000$ J	2+3

	OR: (i) $R \propto l$, $R \propto 1/A$ (ii) $A = \pi r^2 = 3.14 \times (0.25 \times 10^{-3})^2 = 1.96 \times 10^{-7}$ m²; $l = RA/\rho = 10 \times 1.96 \times 10^{-7} / 1.6 \times 10^{-8} \approx 122.7$ m	2+3

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ANSWER KEY & MARKING SCHEME SET C

Chapters 9–12: **Light – Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** goes undeviated	No deviation
2	**(a)** dioptre	m^{-1}
3	**(b)** infinite	Flat surface
4	**(b)** dispersion	Spectrum
5	**(c)** presbyopia	Both near and far
6	**(b)** scattering of light	Molecules
7	**(d)** $H = I^2Rt$	Joule's law
8	**(c)** tungsten	High melting point
9	**(b)** resistance	V/I
10	**(b)** south to north	Closed loops
11	**(a)** current	Forefinger: field
12	**(a)** soft iron core	Temporary magnet

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Speed of light changes in different media.	2
14	Far point: infinity; near point: about 25 cm.	1+1
15	High resistivity and they do not oxidise (burn) readily at high temperatures.	2
16	False — the field at the centre is strong and nearly uniform, perpendicular to the plane of the loop.	2
17	Fuse, MCB (earthing)	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$u = -20$, $f = -15$: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = -\frac{1}{15} + \frac{1}{20} = -\frac{1}{60} \Rightarrow v = -60$ cm; $m = -\frac{v}{u} = -3$; real, inverted, magnified	1+1+1
19	Sunlight enters raindrops; refraction and dispersion; internal reflection; refraction on exit; seen in the direction opposite the Sun	3
20	Cell, key, resistor and ammeter in series; voltmeter in parallel across the resistor	2+1
21	Stretch thumb, forefinger and middle finger mutually perpendicular: forefinger — field, middle finger — current, thumb — force (motion); depends on current and strength of field (and length)	2+1
22	Natural spectrum after rain; water droplets act as prisms: refraction and dispersion on entering, internal reflection, refraction on leaving	1+2
	OR: Light travels a long distance through the atmosphere near the horizon; most blue light is scattered away; red light (longer wavelength) reaches the eye	3
23	Resistance of a conductor of unit length and unit cross-section (Ω m); depends on length, area of cross-section, nature of material (and temperature)	1+2
	OR: $H = I^2Rt = 4 \times 50 \times 10 = 2000$ J	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) twinkling of stars, advanced sunrise/delayed sunset (ii) changing refractive index of the atmosphere (iii) the refractive index of hot air fluctuates	2+2+1
	OR: (i) band of seven colours (ii) different colours travel with different speeds and bend by different angles (iii) two identical prisms, the second inverted (iv) violet	1+2+1+1

25	(i) mains with live, neutral, earth; main fuse, meter, distribution box, parallel circuits (ii) 15 A for high-power appliances (geysers); 5 A for bulbs and fans (iii) earth — safety against leakage; fuse — protection from overloading and short circuits	2+1+2
	OR: (i) field like a bar magnet; uniform inside (ii) place the rod inside the solenoid and pass current (iii) electromagnet: temporary, strength adjustable; permanent: retains magnetism	2+1+2

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